

1. Consider the integral $\int_0^{\pi} x \cos(x) dx$. Which of the following expressions represents this integral as a limit of Riemann sums using right endpoints? Circle the correct answer.

A). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi i}{n} \cos\left(\frac{\pi i}{n}\right) \cdot \frac{\pi i}{n} \right]$

B). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi}{n} \cos\left(\frac{\pi i}{n}\right) \cdot \frac{\pi i}{n} \right]$

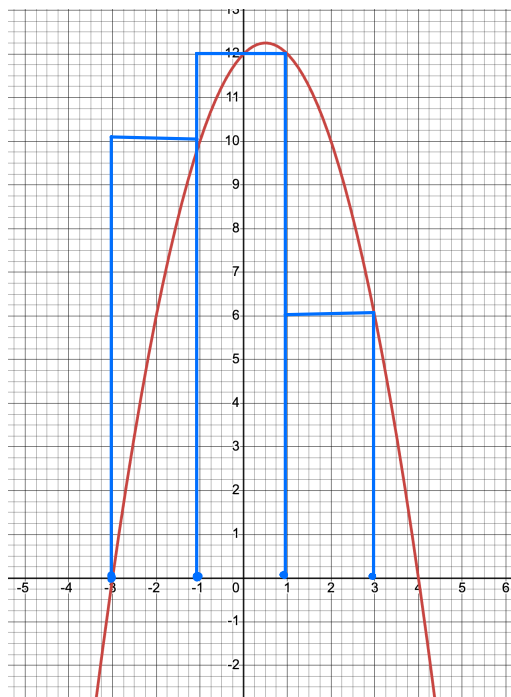
C). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi i}{n} \cos\left(\frac{\pi i}{n}\right) \cdot \frac{\pi}{n} \right]$

D). $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\frac{\pi}{n} \cos\left(\frac{\pi}{n}\right) \cdot \frac{\pi}{n} \right]$

$\Delta x = \frac{\pi - 0}{n} = \frac{\pi}{n}$
 right-endpts: $x_i^* = 0 + i \cdot \Delta x = i \cdot \frac{\pi}{n}$
 $= \frac{\pi i}{n}$
 $f(x_i^*) = \frac{\pi i}{n} \cos\left(\frac{\pi i}{n}\right)$

2. Consider the function $f(x) = 12 - x^2 + x$ on the interval $[-3, 3]$. The graph of the function is given below.

- 1). Sketch the appropriate rectangles on the graph to illustrate the right-endpoint Riemann sum R_3 for $f(x)$ on the interval $[-3, 3]$ with $n = 3$.
- 2). Calculate the right-endpoint Riemann sum R_3 for $f(x)$ on the interval $[-3, 3]$ with $n = 3$.



$\Delta x = 2$

Right-endpts: $-1, 1, 3$

$f(-1) = 12 - 1 - 1 = 10$

$f(1) = 12 - 1 + 1 = 12$

$f(3) = 12 - 9 + 3 = 6$

$R_3 = f(-1) \Delta x + f(1) \Delta x + f(3) \Delta x$

$= 10 \cdot 2 + 12 \cdot 2 + 6 \cdot 2 = 20 + 24 + 12 = 56$